

# Explaining Legal Moves in Equations and Inequalities

## Answer Key

### Algebra 1

#### Quick Answers

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|-------------------|-------------------------------------------|
| 1. 7              | 6. 35                                     |
| 2. (c) = 5, —     | 7. (a) solution set = $(-\infty, -3]$ , — |
| 3. 8              | 8. (b) = 4, —                             |
| 4. $(-4, \infty)$ | 9. $(-\infty, -8)$                        |
| 5. (b) = 6, —     | 10. (a) = 11, —                           |
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#### What is verified

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5 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 5, 7, 8, 10. The worked solution states what a correct response must contain.

The author marked 2 blanks as working space (*not* answers — nothing checks what lands there).

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#### Curriculum

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**Level:** Algebra 1

HSA-REI.A.1 — *problems 2, 5, 7, 8, 10*

HSA-REI.B.3 — *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

*Difficulty 1–5 of 5 across 16 tagged checks.*

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**1.** Balance: left pan  $x + 4$  grams, right pan 11 grams. Complete the model and give  $x$ .

**Solution:** Removing 4 grams from both pans leaves  $x$  on the left and 7 on the right, and the balance is still level — that is exactly what the subtraction property of equality says. The completed model row reads  $x$  and 7.

|         |
|---------|
| $x = 7$ |
|---------|

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**2.**  $3x - 7 = 14 \rightarrow 3x = 21 \rightarrow x = 7$ : (a) name the first move; (b) name the second; (c) solve  $5x + 2 = 27$ .

**Solution (a) — instructor-judged.** 7 was added to both sides: the addition property of equality. On the left  $-7 + 7$  leaves  $3x$ ; on the right  $14 + 7 = 21$ . Full credit states the operation *and* that both sides received it; naming only the number 7 does not.

**Solution (b) — instructor-judged.** Both sides were divided by 3 (equivalently, multiplied by  $\frac{1}{3}$ ): the division property of equality. Full credit again names the operation and both sides.

**Solution (c):** Subtract 2 from both sides:  $5x = 25$ . Divide both sides by 5.

|         |
|---------|
| $x = 5$ |
|---------|

3. Solve  $4(x - 3) = 20$ .

**Solution:** Route one, divide first: dividing both sides by 4 gives  $x - 3 = 5$ , then adding 3 to both sides gives  $x = 8$ . Route two, distribute first:  $4x - 12 = 20$ , add 12 to both sides for  $4x = 32$ , divide both sides by 4. Both routes are legal because each move either does the same thing to both sides or rewrites one side into an equal expression.  $x = 8$

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4. Solve  $-3x < 12$ .

**Solution:** Divide both sides by  $-3$ . Dividing by a negative reverses the order of the two sides, so the  $<$  becomes  $>$ :  $x > -4$ . Checking with a value from the answer set,  $x = 0$  gives  $0 < 12$ , which is true; and  $x = -5$  gives  $15 < 12$ , which is false — so the direction is right.  $x > -4$

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5. Student work  $2x + 5 = 3x - 1 \rightarrow 5 = x - 1 \rightarrow 6 = x$ : (a) name the moves; (b) solve it.

**Solution (a) — instructor-judged.** Step one:  $2x$  was subtracted from both sides (subtraction property of equality), leaving  $5 = x - 1$ . Step two: 1 was added to both sides (addition property), leaving  $6 = x$ . Both moves are legal and the student's work is correct. Full credit names both moves and says each was applied to both sides.

**Solution (b):** Subtracting  $2x$  and then adding 1 to both sides gives  $6 = x$ , which is the same statement as  $x = 6$ .  $x = 6$

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6. Solve  $\frac{x}{5} - 3 = 4$ .

**Solution:** Add 3 to both sides:  $\frac{x}{5} = 7$ . Then multiply both sides by 5:  $x = 35$ . Check:  $35 \div 5 - 3 = 7 - 3 = 4$ .  $x = 35$

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7.  $5 - 2x \geq 11$ : (a) solve; (b) does the sign change, and where?

**Solution (a):** Subtract 5 from both sides:  $-2x \geq 6$ . Divide both sides by  $-2$ , which reverses the sign:  $x \leq -3$ .  $x \leq -3$

**Solution (b) — instructor-judged.** Yes, it changes, and at exactly one step: dividing both sides by  $-2$ . Subtracting 5 from both sides does not affect the direction, because shifting both sides by the same amount preserves which one is larger; multiplying or dividing by a negative swaps which one is larger, so the symbol must be reversed to keep the statement true. Full credit names that step as the reason; a bare yes or no earns half credit.

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8.  $3(x + 2) = 18 \rightarrow 3x + 2 = 18 \rightarrow \dots$ : (a) which step is illegal; (b) solve correctly.

**Solution (a) — instructor-judged.** The very first step.  $3(x + 2)$  was rewritten as  $3x + 2$ : the factor 3 reached the  $x$  but not the 2. The distributive property requires it to multiply every term inside, giving  $3x + 6$ . Note that this is not a property-of-equality error at all — nothing was done to both sides — it is a bad rewrite of one side, which is just as fatal. Full credit points at that first line.

**Solution (b):**  $3(x+2) = 18$  becomes  $3x+6 = 18$ ; subtract 6 from both sides for  $3x = 12$ ; divide both sides by 3.  $x = 4$

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**9.** Solve  $-\frac{x}{4} + 1 > 3$ .

**Solution:** Subtract 1 from both sides:  $-\frac{x}{4} > 2$ . Now multiply both sides by  $-4$ ; the multiplier is negative, so the sign reverses:  $x < -8$ . Check with  $x = -12$ :  $3 + 1 = 4 > 3$ , true.  $x < -8$

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**10.**  $\frac{2x-1}{3} + 4 = x$ : (a) solve; (b) justify each move.

**Solution (a):**

|                          |                                |
|--------------------------|--------------------------------|
| $\frac{2x-1}{3} + 4 = x$ |                                |
| $2x - 1 + 12 = 3x$       | multiply both sides by 3       |
| $2x + 11 = 3x$           | combine like terms on the left |
| $11 = x$                 | subtract $2x$ from both sides  |

Check:  $\frac{2(11)-1}{3} + 4 = \frac{21}{3} + 4 = 7 + 4 = 11$ .

$x = 11$

**Solution (b) — instructor-judged.** Multiplying both sides by 3 is the multiplication property of equality — a both-sides move. Combining  $-1 + 12$  into 11 is a rewrite of the left side only, justified by arithmetic, not by any property of equality. Subtracting  $2x$  from both sides is the subtraction property — another both-sides move. Full credit distinguishes the both-sides moves from the one-side rewrite.